

Comparison of sorghum and maize raw distillates: Factors affecting ethanol efficiency and volatile by-product profile

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ABSTRACT

The potential of sorghum grain in the production of bioethanol was determined in comparison to maize and as an alternative to it. The effects of the raw materials, the raw material cultivars, and the fermentation process on ethanol efficiency and volatile compound profile of the raw distillates were investigated. The ethanol yield proved to be strongly determined by the variety of the raw material and by the fermentation method. The SSF method with a weight to water ratio of 1:4 proved to be the most effective for ethanol production in the case of both maize and sorghum. The highest ethanol yield at 86.94% of the theoretical value, was obtained from maize; the value from sorghum was 80.15%. The fermentation method, though not the grain cultivar, had a significant effect on the volatile compound profile of the raw distillates. The raw distillates obtained using the SSF method at a ground grain to water ratio of 1:4 had the lowest concentration of esters and aldehydes. The average content of volatile compounds in the sorghum distillates was lower than in the maize distillates. This study emphasizes the great potential of sorghum, a maize alternative, as an energy plant in the distillery.

1. Introduction

The production of ethyl alcohol for industrial purposes is an important branch of the economy of many countries. Ethanol is an important raw material in the chemical industry and as solvent with a wide range of applications. This puts pressure on agricultural distilleries to achieve the best possible efficiency and quality of raw spirits, to reduce production costs, and to use increasingly diverse raw materials. In the case of bioethanol production, any improvements that reduce the production costs also increase the competitiveness of ethyl alcohol in relation to other renewable energy sources. Agricultural distilleries produce raw spirits of varying quality through the fermentation of different raw materials and conditions (Pietruszka and Szopa, 2014). The volatile compound profile present in agricultural distillates depends on many factors, including the type of raw materials used, the fermentation and distillation conditions and parameters (Sapińska et al., 2014; Ziółkowska and Jeleń, 2012; Biernacka and Wardencki, 2012; Onuki et al., 2016). The quality of the raw spirits is the main determinant of their subsequent use. Ethanol quality is important for the distilling industry, as it may be used for either food purposes or as a biofuel. When ethanol

is used for other purposes (e.g., as a solvent) impurities are undesirable. The purification of raw spirits increases the total cost of production. It is therefore particularly important to determine the relationship between the parameters of the technological process and volatile compound contents of the raw spirit. In addition to ethanol, raw spirits may contain many volatile compounds as impurities, which can be divided by their chemical properties into basic groups, such as esters, organic acids, carbonyl compounds, and higher alcohols (Ziółkowska and Jeleń, 2012; Wiśniewska et al., 2016). The technical definitions and requirements for ethyl alcohol of agricultural origin are specified in Regulation (EC) No 110/2008 of the European Parliament and of the Council.

Distilling industry is constantly looking for the most suitable raw materials and efficient methods of fermentation in order to achieve the highest possible ethanol yield. The development of the spirits industry is guided by the need to achieve higher levels of efficiency. For this purpose, new raw materials and new technologies are introduced—for example, modifications at the level of starch release and the use of new enzymes. New technological solutions are of particular importance in improving the profitability of ethanol production. Special attention should be paid to fermentation methods based on the simultaneous

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hydrolysis and fermentation of starch, which provide mash with high extract content (very high gravity; VHG) to be fermented, as the glucose is metabolized on a regular basis by yeast cells, which thus avoid osmotic stress, allowing higher ethanol yields to be obtained (Larnaudie et al., 2016; Sapińska et al., 2013). At the same time, this allows investment costs to be reduced and technological operations to be simplified (Strąk and Balcerek, 2015). One important case is optimizing the parameters of the process to the type of raw material being processed. One modification to the simultaneous saccharification and fermentation method is to use an enzymatic preparation such as Stargen (Genencor International), which has the ability to hydrolyze native starch (Szymanowska-Powalowska et al., 2012). This allows to omit the gelatinization and liquefaction steps carried out at high temperatures. However, the degree of conversion of starch granules depends both on the origin of the enzyme and the botanical origin of the substrate itself. Different starch structures make it susceptible to enzymes to varying degrees (Białas et al., 2009; Szymanowska-Powalowska et al., 2012). A very significant distinction is the ratio of amylose to amylopectin for evaluating enzymatic hydrolysis and further fermentability of the starch raw material. Increasing amylose content reduces the hydrolysis of native starch by amylases (Tester et al., 2006). The size of the starch granules may result in its resistance to α -amylase and affects the susceptibility to hydrolysis (Knutson et al., 1982; Rendleman and Jacob, 2000).

Maize (*Zea mays* L.) is one of the most important raw materials for energy purposes. In recent years, the use of other parts of this plant has been of great interest. Maize exceeds other raw materials used for the production of bioethanol in starch content (over 72%, dry basis), harvest yield, and ethanol yield (Zabed et al., 2017). According to FAO data, maize was grown on 17 536.535 thousands ha in Europe with a yield of 6.29 tons per ha (Food and Agriculture Organization of the United Nations, 2017).

Increasingly, however, during the flowering and pollination of maize, there are long-lasting droughts, which not only result in a drastic drop in yield, but also a deterioration in quality. In order to efficiently utilize production potential, raw materials are sought that can be used for cultivation under such conditions, replacing other plants. Great hopes are attached to sorghum (*Sorghum* Moench), which is a close relative of maize, and which thus has many similar features in both appearance, developmental physiology, and applications (Nghiem et al., 2016; Mathur et al., 2017). Sorghum has tolerance to drought and needs less fertilizer than maize. As a C4 photosynthesis plant, it has a high efficiency in using solar radiation and heat, therefore, it manages water much more efficiently than maize. These properties and the occurrence of rainless periods leads many farmers to grow this crop under dry conditions instead of maize. Sorghum is also characterized by high starch (64.3–73.8%, dry basis) and protein (8.19–14.02%, dry basis) content (Nghiem et al., 2016). Sorghum is the cereal crop planted also in Europe on an area of 295.853 thousands ha and yielding 3.19 tons per ha (Food and Agriculture Organization of the United Nations, 2017).

No data on the volatile compounds of spirits obtained from grain sorghum were found in the available literature, which mostly describes the use of analytical techniques to analyze distillates of agricultural origin (Onuki et al., 2016; Wiśniewska et al., 2016) and to identify spirits of various origins (Biernacka and Wardencki, 2012; Jeleń et al., 2010; Satora et al., 2017). The requirements for spirits include factors indicating that specific conditions and process parameters should be used to obtain alcohol beverages of certain quality. Chemical impurities in raw spirits are very significant in the production of bioethanol and of food, because the quality requirements determine the allowed ranges of groups of chemical compounds. The main objective of this study was thus to quantify the contamination of distillates arising from different technological processes using grain sorghum and maize as raw fermentation materials. We present detailed profiles of the volatile by-products created during ethanol fermentations. An important part of the work is its demonstration of the potential of sorghum as a raw

material in the distillation industry, which can serve as a plant alternative to maize that is capable of yielding under European climate conditions.

2. Methodology

2.1. Raw material

Sorghum (*Sorghum* Moench) and maize (*Zea mays* L.) were used in this research as fermentation materials (not genetically modified). Five cultivars of sorghum (GK 813, GK J-018, Sweet Susana, Sweet Caroline, and GK Emese) were employed. The experimental plots, harvested in 2017, were located at the Stary Sielec Experimental Station in Poland, which belongs to the Institute of Natural Fibers and Medicinal Plants. The maize cultivars were supplied by the IHAR Plant Breeding Group in Smolice and were planted in 2017. The five tested cultivars of maize were Rosomak, SM Hubal, Fortop, SM Kurant, and SM Furman.

2.2. Microorganisms

Two distillery yeast strains (*Saccharomyces cerevisiae*) recommended for the selected fermentation methods were used in the study. In the separate hydrolysis and fermentation method (SHF), *Saccharomyces cerevisiae* in the form of a dried Fermiol formulation (Lesaffre, France) was used. In the simultaneous saccharification and fermentation method (SSF), the Ethanol Red strain was used for ethanol production (Lesaffre, France); this is especially recommended for higher gravity mash. Although different mash concentrations were used in SSF method, the same yeast strain was applied for later comparison. The dry yeast was hydrated before use and the slurry was added to the mash (0.5 g of dried yeast/L).

2.3. Enzymes

Four enzyme preparations were used in the SHF process: Termamyl SC DS (thermostable α -amylase from *Bacillus licheniformis*; Novozymes, Denmark) was used in the liquefaction of the ground grain, and San Extra L (glucoamylase from *Aspergillus niger*; Novozymes, Denmark) was used for starch saccharification. Additionally, Optimash VR (a complex of xylanase and cellulase from *Penicillium funiculosum*; Genencor International, USA) was used to decompose the nonstarchy polysaccharides, and GC 106 Protease (from *Aspergillus niger*; Genencor International, USA) hydrolyzed the proteins present in the flour.

The SSF process was conducted using bacterial α -amylase Amylex BT2 (Genencor International, USA) produced by fermentation of a genetically unmodified strain of *Bacillus stearothermophilus*, for starch liquefaction. Otherwise, Diazyme SSF containing glucoamylase and protease (both derived from *Aspergillus niger*; Genencor International, USA) was used for starch hydrolysis.

Stargen 001 (a granular starch hydrolyzing enzyme containing α -amylase from *Aspergillus kawachi* and glucoamylase from *Aspergillus niger* (Genencor International, USA)) was used in SSF for the uncooked and unliquefied starch. Additionally, Optimash VR, and GC 106 Protease were used in this process.

All enzymes were used according to the producers' recommendations.

2.4. Fermentation processes

The grain samples were ground prior to analysis on a WZ-1 laboratory mill (meal size < 0.8 mm) and the volume of the fermentation media was 150 ml. Distillation was carried out after 72 h of fermentation. The entire fermentation medium was distilled.

SHF method. The ground grain was mixed with water at a 1:6 ratio, and the pH of the fermentation media was adjusted to 5.0–5.5. Non-pressure cooking (100 °C, 1 h) was used to gelatinize the starch. Next,

liquefaction (80 °C, 20 min) with Termamyl SC DS (0.2 mL/kg of grain) and saccharification (55–60 °C, 100 min) with San Extra L (0.6 mL/kg of grain), Optimash VR (0.08 mL/kg of grain), and GC 106 Protease (0.08 mL/kg of grain) were performed. After hydrolysis, the fermentation media were supplemented with diammonium phosphate (0.4 g/L) and inoculated with *S. cerevisiae* (Fermiol). The samples were incubated at 30 °C for 72 h (Szambelan et al., 2018a, 2018b).

SSF method. The ground grain samples were mixed with water at ratios of 1:2, 1:3, and 1:4. After adjusting the pH of the media to 5.0, Amylex BT2 (0.8 mL/kg of grain) was added. The liquefaction process was carried out at 95 °C for 45 min. The liquefied mash was cooled to 35 °C and Diazyme SSF (0.075% of the weight of substrate) was added one hour before inoculation with Ethanol Red distillery yeast. The media were fermented for 72 h at 35 °C and 125 rpm (Szambelan et al., 2018a, 2018b).

SSF method with Stargen 001. From the resulting meal, a suspension with water was prepared in a 1:3 ratio. The pH was then adjusted to 4.5 and Stargen 001, Optimash VR, and GC 106 Protease were added; the medium was incubated at 35 °C for 1 h. The fermentation medium was then supplemented with diammonium phosphate (0.4 g/L) and inoculated with *S. cerevisiae*, Ethanol Red strain (0.5 g/L). The samples were incubated at 35 °C for 72 h with constant stirring at 125 rpm (Szambelan et al., 2018a, 2018b). In the case of sorghum, before the addition of the enzyme, liquefaction was performed at 65 °C for 1 h with the addition of α -amylase Amylex BT2 (0.8 mL/kg of grain) (Nkombe et al., 2016). The liquefaction process was used because otherwise the SSF process with Stargen 001 resulted in very low ethanol yields from sorghum. The Stargen 001 enzyme is effective specifically on native maize starch, and the molecular structure of starch in other raw materials can drastically limit its hydrolysis efficiency.

The raw distillates thus obtained were investigated for their levels of ethanol and volatile by-products. The remaining reducing sugars were measured in the stillage.

2.5. Analytical methods

The reducing sugars in the grains and stillage were determined using the 3,5-dinitrosalicylic acid method (Miller, 1959). The starch content was determined using the method of Hölm et al. (1986). Protein content was estimated as total nitrogen by the Kjeldahl method (Krelowska-Kulas, 1993). The ethanol was distilled from the fermented media using the Super Dee digital stiller (Gibertini, Italy). The ethanol concentration of the raw distillates was measured on the DDM 2911 Automatic Density Meter (Rudolph Research Analytical, USA). The total acidity of the raw distillates was determined by colorimetric method and expressed in grams of acetic acid per liter of 100% ethanol (ISO 1388/2, 1981).

2.6. Gas chromatography for volatile compounds analysis

In order to estimate the profile of the volatile compounds, an HP 6890 equipped with split/splitless inlet, two flame ionization detectors (FID), and autosampler were used to analyze the resulting raw distillates. The compounds were separating using two capillary columns: column A: Agilent CP-Wax 57CB (60 m $\times$ 0.25 mm $\times$ 0.4 μ m); column B: Agilent DB-624 (60 m $\times$ 0.25 mm $\times$ 1.4 μ m) linked with the injection port using a Restek inlet adaptor kit for dual column installation (cat.# 27185). The length and diameter of both columns was equal, to facilitate the same flow. As determined from the ratio of peak areas of the compounds eluting from columns A and B, the A/B column split was 1.19 (average for all compounds analyzed on both columns).

Volatile compounds were calculated based on the internal standard 4-methylpentan-2-ol (5 μ l of ethanol solution (0.25 g/mL)), which was added to each spirit sample in a 25 mL volumetric flask until the concentration of the internal standard was 50 mg/L. GC autosampler vials contained approximately 1.5 mL of the sample, of which 2 μ l was

injected simultaneously into two capillary columns in split (1:20) mode. The temperature for vaporization was 220 °C. Compounds were resolved at an initial programmed temperature of 35 °C for 4 min; the temperature was then increased to 95 °C at the temperature gradient of 20 °C/min, held 5 min, and afterward increased to 190 °C at the temperature gradient of 10 °C/min, where they were held 3.25 min. The entire analysis run lasted 24.75 min. Hydrogen was the carrier gas, with the flow set on the injection port at 4.1 mL/min in constant flow mode; the initial pressure was 30 MPa. Two detectors were set up with the same parameters; the combustion temperature was 280 °C and was supplied by H₂ flow (30 mL/min), air flow (300 mL/min), and N₂ makeup flow (5 mL/min). The acquisition rate was 20 Hz. The compounds were quantified using the internal standard method (ISTD) with ChemStation Rev. B.04.03-SP2. The correlation coefficients (R^2) for all standard curves were high and ranged from 0.9977 to 1.0000 for all analytes. The precision of the method was estimated with low RSD ranging from 0.19% to 13.91%.

2.7. Statistical analysis

All the experiments were carried out in three independent repetitions. The samples were compared using multivariate statistics as k-means, cluster analysis, and principal component analysis (PCA). The Mann–Whitney *U* test was used to show significant differences between two mean values. The Kruskal–Wallis ANOVA by ranks test, and multiple comparisons of mean ranks for groups (Dunn's test), were carried out when more than two groups were being compared. The Spearman rank correlation coefficient was calculated to assess the relationship between two variables. The statistical hypothesis were tested at significance level $\alpha = 0.05$. The statistical analysis was performed using Statistica software, Version 10, StatSoft Inc., OK-US.

3. Results and discussion

3.1. Characteristics of sorghum and maize cultivars

The cultivars of sorghum and maize used in this research were characterized in terms of their starch, reducing sugar, and protein contents. ANOVA statistical analysis showed that the cultivars of both sorghum and maize significantly ($p < 0.05$) differed in starch content. The sorghum cultivars were characterized by high content of starch: GK J-018 61.77%, GK Emese 62.23%, Sweet Susana 62.55%, GK 813 65.37%, and Sweet Caroline 67.18%. Starch content in the maize cultivars was 59.44% for SM Furman, 60.76% for SM Kurant, 60.88% for Rosomak, 62.92% for Fortop, and 63.29% for SM Hubal. Total protein content varied from 9.05 to 10.41% in maize grain, and from 9.22 to 11.19% in sorghum grain.

3.2. Separate hydrolysis and fermentation and simultaneous saccharification and fermentation efficiency

The separate hydrolysis and fermentation (SHF), as well as the SSF process involving several density variants and SSF with Stargen 001, were used to determine the proper conditions for the production of raw distillates with the expected levels of volatile by-products. The ethanol yields, sugar in stillage, and total acidity of spirits from sorghum and maize cultivars are presented in Table 1. It can be concluded from our results that the simultaneous saccharification and fermentation (SSF) process with a mash concentration of 1:4 was the most efficient. ANOVA confirmed this conclusion for the sorghum and maize cultivars (Table 1). However, unlike the starch content, the average ethanol yield from sorghum was lower than from maize. The concentration of sugars in stillage and the total acidity of raw distillates were not determined by the fermentation method alone. A more detailed comparison was performed using the k-means algorithm to generate homogenous groups for maize and sorghum cultivars using the parameters before and after

Table 1

Parameters compilation after fermentation: efficiency of ethanol production, sugar in stillage and total acidity of raw distillates from maize and sorghum grain depending on the type of fermentation method.

Grain	Maize				
Method*	SHF	SSF (1:4)	SSF (1:3)	SSF (1:2)	SSF (1:3) + Stargen
Parameter	Ethanol (L/100 kg starch)				
Mean	55.87 ^{cB}	60.19 ^{dB}	57.44 ^{cB}	47.96 ^{bB}	41.49 ^{aA}
SD	1.61	2.30	2.13	2.41	4.09
Parameter	Ethanol (L/100 kg grain)				
Mean	34.32 ^{cB}	36.96 ^{dA}	35.29 ^{cA}	29.45 ^{bA}	25.50 ^{aA}
SD	0.83	0.76	1.02	1.10	2.63
Parameter	Sugars in stillage (mg/g)				
Mean	5.07 ^{bB}	1.47 ^{aB}	5.86 ^{bB}	1.01 ^{aA}	1.92 ^{aA}
SD	1.60	0.11	2.58	0.17	0.37
Parameter	Total acidity of spirits (g/L 100% EtOH)				
Mean	0.04 ^{aA}	0.06 ^{dA}	0.05 ^{bA}	0.06 ^{dA}	0.05 ^{abA}
SD	0.01	0.02	0.01	0.01	0.01
Parameter	Ethanol (% v/v)				
Mean	8.60 ^{aB}	11.09 ^{cB}	13.25 ^{dB}	14.73 ^{cA}	9.56 ^{bA}
SD	0.21	0.23	0.39	0.56	1.00
Grain	Sorghum				
Method	SHF	SSF (1:4)	SSF (1:3)	SSF (1:2)	SSF (1:3) + Stargen
Parameter	Ethanol (L/100 kg starch)				
Mean	48.85 ^{bA}	55.99 ^{dA}	53.62 ^{cA}	45.40 ^{aA}	45.95 ^{aB}
SD	1.97	1.25	1.84	2.98	5.28
Parameter	Ethanol (L/100 kg grain)				
Mean	31.16 ^{bA}	35.75 ^{cA}	34.24 ^{cA}	28.98 ^{aA}	29.41 ^{abA}
SD	1.34	1.85	2.10	2.18	4.21
Parameter	Sugars in stillage (mg/g)				
Mean	2.12 ^{bA}	1.37 ^{aA}	2.40 ^{bA}	8.65 ^{dB}	3.42 ^{cB}
SD	0.28	0.17	0.49	1.68	0.55
Parameter	Total acidity of spirits (g/L 100% EtOH)				
Mean	0.10 ^{cB}	0.08 ^{bB}	0.09 ^{cB}	0.06 ^{aA}	0.08 ^{bB}
SD	0.01	0.01	0.01	0.01	0.01
Parameter	Ethanol (% v/v)				
Mean	7.79 ^{aA}	10.72 ^{aA}	12.84 ^{cA}	14.49 ^{dA}	11.03 ^{bB}
SD	0.33	0.55	0.79	1.09	1.58

*fermentation method: SHF – separate hydrolysis and fermentation; SSF – simultaneous saccharification and fermentation.

^{a-c}Values with different letters within the same row are significantly different ($p < 0.05$).

^{A, B}Values with different letters within the same column/parameter are significantly different ($p < 0.05$).

fermentation. Three groups were generated for the maize and sorghum samples. The *k*-means profiles are presented in Fig. 1. The most discriminative parameter for maize (*F*-value 493.31) was sugars in grain; the concentration of starch in grain was one of the less discriminative variables (*F*-value 9.45). The first and third groups have the same grain parameters but totally different ethanol efficiencies; Fig. 1 and Table 2 show that the first group contains most of the samples fermented using the SSF (1:2) and SSF (1:3) + Stargen methods. The third group contains samples fermented with SHF, SSF (1:3), and SSF (1:4). This demonstrated that the choice of fermentation method can dramatically alter fermentation efficiency, but that in the case of the Rosomak cultivar, the fermentation method does not have a great effect on the ethanol efficiency, as indicated by the *k*-means clustering results; all samples are in the second group. Nevertheless, detailed comparison shows that it is possible to distinguish the Rosomak maize sample after fermentation; the SSF method (1:4) was the most effective, and SSF (1:3) + Stargen method was the least effective.

A similar comparison was performed for the sorghum samples. The most discriminative parameter for sorghum (*F*-value 491.4) was starch in the grain; the total acidity of spirits and the total protein in grain were less discriminative variables (*F*-value 9.6 and 20.2, respectively). The

sorghum cultivars GK 813 and Sweet Caroline were classified into the third group by the *k*-means algorithm (Table 2), independently of fermentation method. This group has following *k*-means profile (Fig. 1B): high starch content, low protein, low sugar content in grain, and high fermentation efficiency. A detailed comparison showed that the SSF (1:4) method was again the most effective here. The first cluster contained sorghum samples for which the SSF (1:2) and SSF (1:3) + Stargen methods were particularly unfavorable. The sorghum samples in the second group had high ethanol efficiency despite having a starch content similar to that of the third group.

Comparing the efficiency of fermentation per mass of grain (Fig. 1), it was noticed that it is possible to obtain the same high ethanol yield for the maize and sorghum samples in the third cluster. However, the sorghum samples have higher potential due to their higher starch concentration in the grain, while the differences between sugars in the grain were statistically insignificant. We additionally noted the advantage possessed by the sorghum samples in this case, since the third cluster consisted of cultivars that are not as susceptible to the fermentation method. In summary, the highest ethanol yields were obtained for GK 813 sorghum and SM Furman maize, at 80.15% and 86.94% of the theoretical yield, respectively. The ethanol yields for the SSF process showed a significant increase ($p < 0.05$) for both sorghum and maize cultivars when the density of the mashes decreased.

The ethanol yields of the SSF process with the Stargen 001 enzyme were found to be lowest for both sorghum and maize. One reason for this result may be the weather conditions during the growing season, which have a large effect on the efficiency of the process: from the moment of sowing (soil temperature), through the growth stage (the plant's water needed) to the time of grain harvest (limited precipitation). During cultivation in 2017, the seeds, due to prolonged cold weather, were sown in unheated soil and grew only slowly. During the growth and development of the plant, there was insufficient rainfall. Harvest time was delayed due to unfavorable weather in the form of abundant rainfall. The increased moisture content of the grain favors microbiological infections. This may be the reason for the low effectiveness of the SSF variant with Stargen 001, which gave better results from the plants harvested in 2016 (Szambelan et al., 2018a, 2018b), as this method involves no heat treatment that could help reduce the risk of infection. In the case of sorghum, which was also harvested late from the field, the SSF method with Stargen 001 gave negligible fermentation yields. For this reason, we used initial liquefaction of the raw material (see Section 2.4, *Fermentation processes*). The modified fermentation conditions allowed for better results of 52.90%–72.54% of the theoretical ethanol yield, while Nkombe et al. (2016) obtained 51.8%–78.8%. In general, the efficiency of the method using Stargen 001 is strongly related to the quality of the raw material. This method should undergo further research in order to achieve higher performance parameters.

These results also indicate that fermentation efficiency requires integrated knowledge of the correlation of starch content and ethanol yield, as well as regarding the quality of the raw material.

3.3. Analysis of the volatile profile of the raw distillates

The main stage of the research was the analysis of the raw distillates produced from sorghum and maize. Significant differences were noted between the mean values of the total volatile compounds in the maize (7104.22 mg/L) and sorghum (6757.80 mg/L) samples. However, these data do not take into account the differences due to the cultivars and fermentation methods. The quantitative composition of the raw distillates was analyzed depending on the fermentation parameters as well as the raw material used and its cultivar.

Cluster analysis was performed to determine the similarity of volatile compound profiles between the samples (Fig. 2). The samples obtained using the SHF method were similar to those samples fermented by the SSF method with Stargen 001. The maize and sorghum samples obtained using SSF with a mash concentration of 1:2 were close to each other on

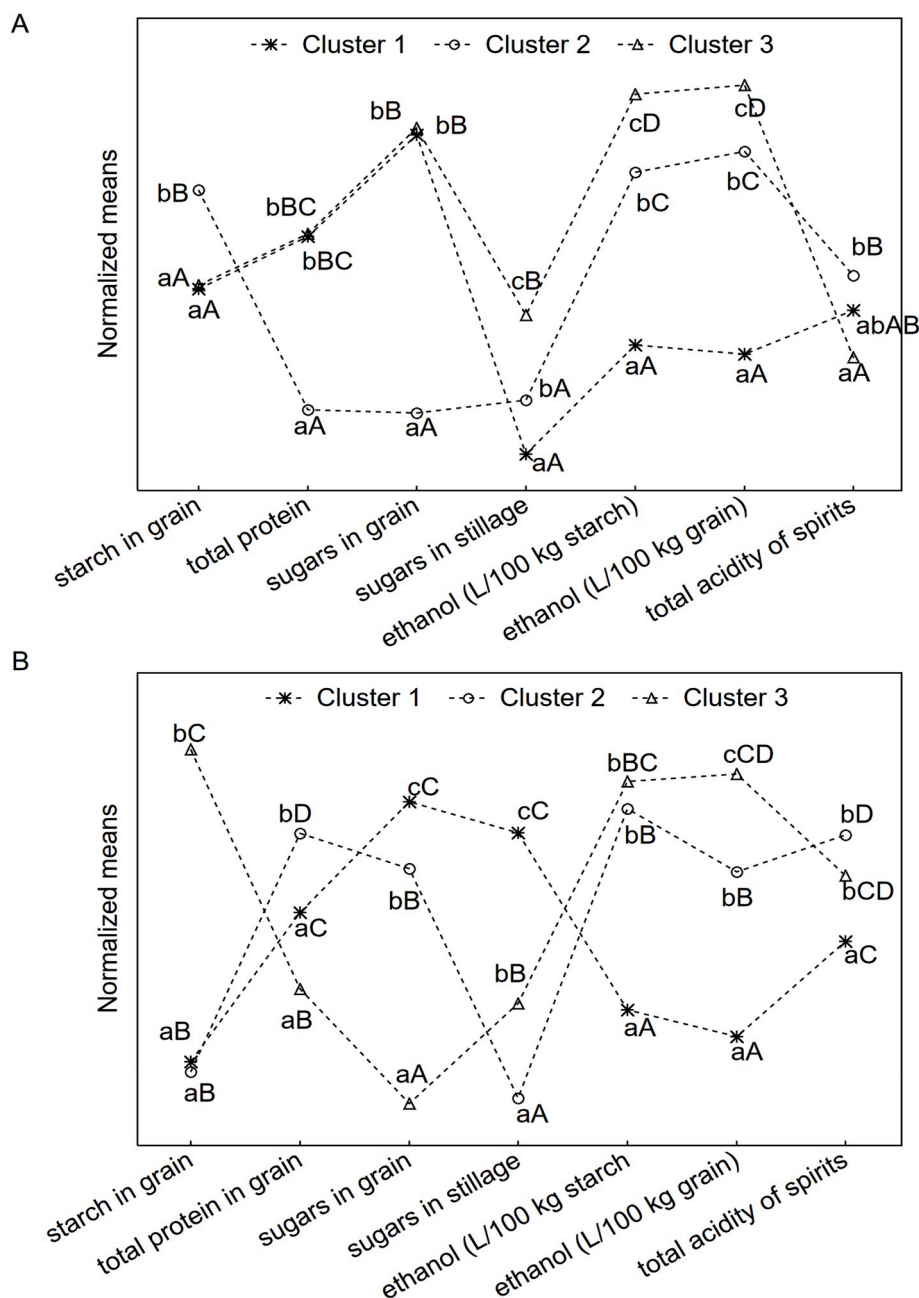


Fig. 1. Graph of means, clusters were calculated separately for maize (A) and sorghum (B) samples. The clusters with different lower-case (^{a-c}) within maize or sorghum samples indicate significantly different mean values ($p < 0.05$). The different upper-case (^{A-D}) within maize and sorghum samples indicate different mean values ($p < 0.05$) for all six clusters. Mean values grow according to order of the alphabet.

the dendrogram, and the same was true for the 1:4 mash concentration variant. This suggests that the fermentation method has a much greater effect than the grain variety on the volatile compound profile of the raw distillates.

PCA was performed to describe the relation between the volatile compounds, fermentation method, and the cultivars of maize and sorghum grain. The PCA (principal component analysis) model explains 44% of the total variance and shows a clear pattern of variables and samples in Fig. 3A and Fig. 3B, respectively. No difference was seen between the volatile profiles for the maize cultivars within each fermentation method. The sorghum samples also for homogenic groups by fermentation method. The samples were systematically arranged along the PC2 axis, with the maize samples having lower values of PC2 than the sorghum samples. However, the large shift along the PC1 axis demonstrates the significant effect of the fermentation method.

Comparing the loading and score plot, it can be seen that the raw distillates obtained using the SSF 1:2 method contain high concentrations of esters and aldehydes. A reduction in aldehydes and esters was observed for the SSF method as the ratio of ground grain to water increased from 1:2 to 1:4. Aldehydes were the most significant group of volatile compounds detected. Of the aldehydes, acetaldehyde was present in the largest amount in the distillates, but these quantities were lower (except for SSF 1:2), than in the study of Sapińska et al. (2014) for maize spirits. In addition to acetaldehyde, 2-methylpropanal—which reduces the sensory quality of spirits—was identified in small quantities. However, it is worth noting that smaller amounts ($p < 0.05$) of this compound were found in the sorghum distillates than in the maize distillates. While Sapińska et al. (2014) did not detect 2-methylpropanal in maize distillates, they did find it in rye spirit. The quantity of these aldehydes exceeded the permitted standards for raw spirits of

Table 2

K-means clustering results, cluster members were calculated separately for maize and sorghum^a samples. Number of clusters generated for maize and sorghum: 3 and 3 respectively.

Grain	Method ^a				
Maize:	SHF	SSF (1:3)	SSF (1:4)	SSF (1:3) + Stargen	SSF (1:2)
Rosomak	2	2	2	2	2
SM Hubal	3	3	2	1	2
SM Kurant	3	3	3	1	1
SM Furman	3	3	3	1	1
Fortop	3	3	3	1	1
Sorghum:	SHF	SSF (1:3)	SSF (1:4)	SSF (1:3) + Stargen	SSF (1:2)
Sweet Susana	2 ^a	2 ^a	2 ^a	1 ^a	1 ^a
GK Emese	2 ^a	2 ^a	2 ^a	2 ^a	1 ^a
GK J-018	2 ^a	2 ^a	2 ^a	2 ^a	1 ^a
GK 813	3 ^a	3 ^a	3 ^a	3 ^a	3 ^a
Sweet Caroline	3 ^a	3 ^a	3 ^a	3 ^a	3 ^a

^a Fermentation method: SHF – separate hydrolysis and fermentation; SSF – simultaneous saccharification and fermentation.

agricultural origin. However, it should be noted that, in this study, the volatile compound levels were determined in the crude distillates, which are only later concentrated in order to obtain raw agricultural spirits. Aldehydes are a fraction that contributes to spirits a very unpleasant odor and taste, but a properly conducted rectification and refining processes could separate them from the ethanol (Christoph and Bauer-Christoph, 2007). Thus, from the theoretical and practical points of view, there is no substantive basis for disqualifying raw distillates as a raw material only because the amount of aldehydes contained in them exceeding the permitted level. The most frequently occurring esters in our raw distillates were ethyl acetate and propyl acetate. The

concentration of ethyl acetate was lower than indicated by Sapińska et al. (2014) for maize distillates in both our sorghum and maize distillates, except in the case of samples made using SSF with Stargen 001. As in the case of aldehydes, the level of esters in the raw distillates increased with the mash density in the SSF process. Analysis of these compounds allowed us to observe that, as the mash density increased in the SSF method, the amount of ethyl acetate decreased, while the level of propyl acetate significantly increased ($p < 0.05$).

The highest concentrations of methanol and higher alcohols were detected in samples from the SHF and SSF with Stargen 001 processes, respectively. Higher alcohols, as fermentation by-products, are an important group from the quantitative point of view. The amount of higher alcohols that formed as a result of the deamination of amino acids (and metabolism of carbohydrates and other nitrogen-free compounds) in our raw distillates exceeded Polish standards (2002) for spirits of agricultural origin (the minimum concentration of ethanol in the raw spirit of cereal origin is 88% v/v), irrespective of the raw material and the fermentation method. It should be noted that the total amount of higher alcohols in the raw distillates obtained using the SSF method was lower ($p < 0.05$) when the ratio of the ground grain to water was 1:2, in the case of both maize and sorghum. The highest concentration ($p < 0.05$) of higher alcohols was found in maize and sorghum samples from the SHF method. The predominant higher alcohols in the distillates proved to be 3-methylbutan-1-ol, 2-methylbutan-1-ol, and 2-methylpropan-1-ol. 3-methylbutan-1-ol and 2-methylbutan-1-ol are characterized by a malt flavor, and were found in much smaller quantities in the SSF distillates than in the SHF distillates, and in much smaller quantities in the sorghum samples than in the maize samples. The distillates also contained propan-1-ol, the flavor quality of which is referred to as stupefying (Christoph and Bauer-Christoph, 2007). It was noted that twice as little of this compound was detected in the distillates from that SSF method as compared to those from the SHF and SSF with Stargen 001 method, for both maize and sorghum distillates. The higher alcohols in

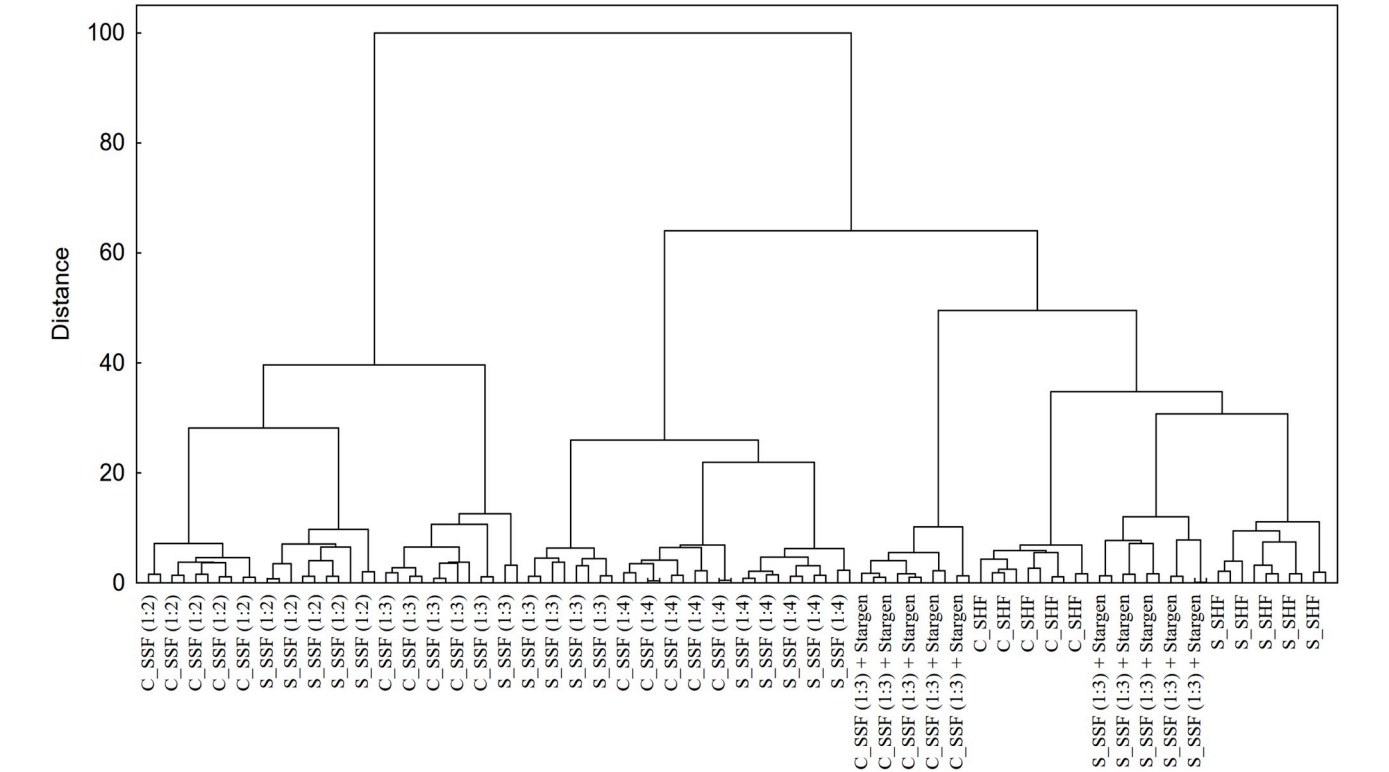


Fig. 2. Cluster analysis, similarity of distillates according to volatile compounds. Samples obtained after fermentation of C – maize and S – sorghum, SHF – separate hydrolysis and fermentation, SSF – simultaneous saccharification and fermentation of starch; the ground grain samples were mixed with water in a 1 : 2, 1 : 3 and 1 : 4 ratio.

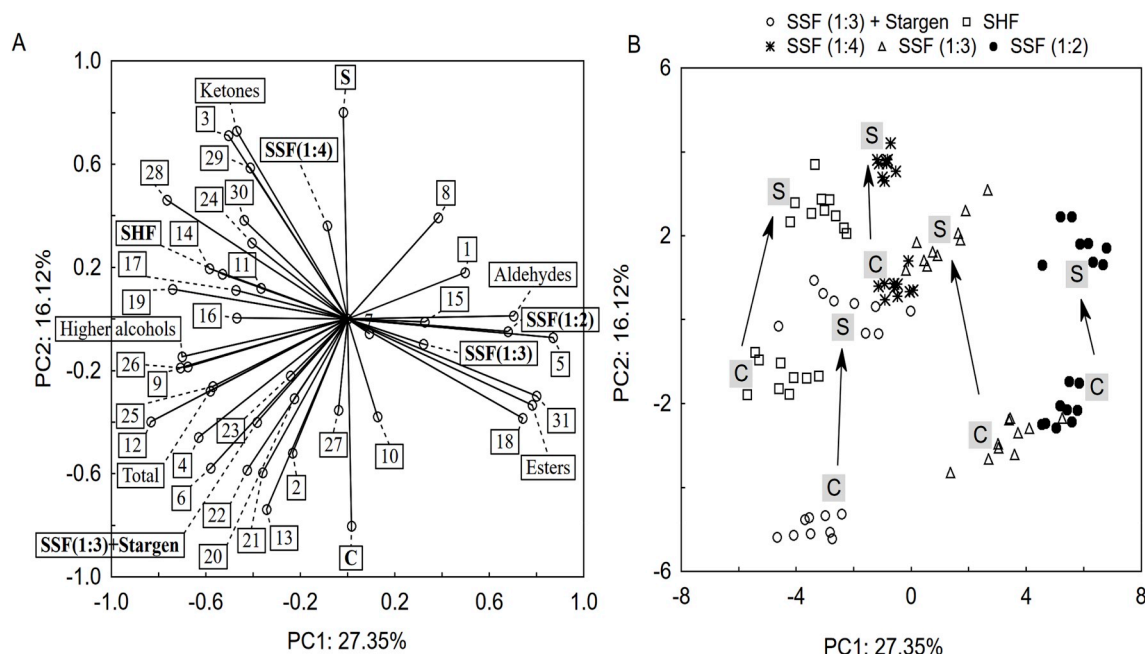


Fig. 3. PCA, A – loading plot: 1 – acetaldehyde, 2 – isobutyraldehyde, 3 – acetone, 4 – ethyl acetate, 5 – acetal, 6 – methanol, 7 – ethyl butyrate, 8 – 2-butanol, 9 – 1-propanol, 10 – butyl acetate, 11 – hexanal, 12 – isobutanol, 13 – isoamyl acetate, 14 – ethyl pentanoate, 15 – 1-butanol, 16 – 2-methyl-1-butanol, 17 – 3-methyl-1-butanol, 18 – 2-hexanol, 19 – 1-pentanol, 20 – 1-hexanol, 21 – ethyl octanoate, 22 – 1-heptanol, 23 – furfural, 24 – benzaldehyde, 25 – 1-octanol, 26 – ethyl decanoate, 27 – phenylacetaldehyde, 28 – 2-phenylethyl acetate, 29 – 2-phenylethanol, 30 – pentanal, 31 – propyl acetate, aldehydes, esters, higher alcohols, ketones, total; B – score plot: C – corn and S – sorghum, SHF – separate hydrolysis and fermentation, SSF – simultaneous saccharification and fermentation of starch; the ground grain samples were mixed with water in a 1 : 2, 1 : 3 and 1 : 4 ratio.

excessive amounts are a group of volatile by-products that degrade the quality of agricultural raw spirits and are undesirable in the production of pure vodkas. They are a valuable component of the flavor bouquet of wine, cognac or beer. However, if the raw spirit is intended for production of highly concentrated alcohols, higher alcohols are almost entirely removed during rectification process (Christoph and Bauer-Christoph, 2007; Sapińska et al., 2014).

The raw distillates also contained methanol. Analysis of the sorghum and maize raw distillates showed that a lower quantity of methanol ($p < 0.05$) was detected in the case of the SSF method than in the case of SHF or SSF with Stargen 001 (Table 3). It was also noted that the presence of methanol depended on the raw material. Studies have shown that distillates from sorghum are characterized by lower methanol content ($p < 0.05$) within the method than are distillates from maize. The values of methanol detected fulfilled the requirements for methanol content in agricultural spirits established in Polish regulations (PN) (2002) and in European Regulations (EC) No 110/2008 (Regulation (EC), 1981).

Moderate levels of methanol, aldehydes, ketones, and esters were

found in samples made using the SSF 1:4 method; however, the lowest concentration of higher alcohols occurred using SSF 1:2. These observations were confirmed using the Kruskal–Wallis ANOVA by ranks test and post-hoc comparisons (Table 3).

The results show that both the raw materials used and the fermentation method have great effects on the quality and composition of the volatile profile in raw distillates. The levels of the residues detected in the raw distillates here were not in line with Polish or European regulations (EC) No. 110/2008 (Regulation (EC), 1981). However, it should be recalled that the requirements concern ethyl alcohol of agricultural origin, not the raw distillate. This study provides new data on the creation of the volatile compound profile of distillates by the introduction of new raw materials and the use of fermentation process parameters adapted to the raw material and its cultivar.

4. Conclusion

This research points to sorghum as a potential raw material for the production of ethanol, with particular attention to the quality of the

Table 3

Compilation of volatile compounds (mg/L 100% EtOH) according to fermentation method (SHF – separate hydrolysis and fermentation, SSF – simultaneous saccharification and fermentation of starch; the ground grain samples were mixed with water in a 1 : 2, 1 : 3, and 1 : 4 ratio).

Grain	Method	Methanol	Aldehydes	Ketones	Esters	Higher alcohols
Maize	SHF	50.46 ^{bc}	179.19 ^{ab}	10.09 ^b	158.36 ^{ab}	8179.88 ^e
Maize	SSF (1:4)	40.68 ^{abc}	134.19 ^a	10.01 ^b	89.56 ^{ab}	5529.44 ^{bc}
Maize	SSF (1:3)	49.40 ^{bc}	351.27 ^c	7.10 ^a	306.89 ^c	6906.87 ^d
Maize	SSF (1:2)	38.25 ^{ab}	670.09 ^d	7.95 ^a	513.08 ^d	4766.72 ^{ab}
Maize	SSF (1:3) + Stargen	194.40 ^e	154.74 ^{ab}	7.78 ^a	127.63 ^{ab}	6262.81 ^{cd}
Sorghum	SHF	54.16 ^c	284.04 ^{bc}	14.62 ^c	57.79 ^a	8178.77 ^e
Sorghum	SSF (1:4)	35.95 ^{ab}	171.56 ^{ab}	11.87 ^c	95.75 ^{ab}	5239.24 ^{bc}
Sorghum	SSF (1:3)	33.41 ^a	250.16 ^{abc}	9.39 ^b	178.59 ^b	6630.34 ^d
Sorghum	SSF (1:2)	32.92 ^a	671.73 ^d	9.53 ^b	486.42 ^d	4011.80 ^a
Sorghum	SSF (1:3) + Stargen	71.09 ^d	377.70 ^c	13.28 ^d	193.06 ^b	6083.70 ^{cd}
Polish standards for raw spirit		<80	<100	–	–	<3500

^{a–e} Values with different letters within the same column are significantly different ($p < 0.05$).

distillate. After a series of studies on the performance of alcoholic fermentation, it was observed that, for both sorghum and maize, the highest ethanol yield can be obtained using the SSF process with a mash concentration of 1:4. Some sorghum cultivars were characterized by the same efficiency of fermentation per mass of grain, like some maize cultivars. The next stage of this study demonstrated that raw distillates from grain sorghum and maize were characterized by lower quantities of total volatile by-products when the SSF method was used than in the case of the classic technological process, SHF. This research into SSF leads to the conclusion that the density of the mash has a significant effect on the quality of the raw distillates. Increasing the mash density led to an increase in the amount of aldehydes and esters. It has been demonstrated that ethanol yield is determined by both the cultivar and the method, while the volatile fermentation compounds are determined mainly by the fermentation method. Sorghum can be considered as an alternative species to maize that can be grown in areas where the growth of maize is limited by soil or weather conditions. To confirm the conclusions of the study, the results should be tested on a larger scale.

Declaration of competing interest

There are no conflicts of interest to declare.

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